X-Fusion: A Tri-Modal Cross-Attention Vision Transformer with Contrastive Alignment for Interpretable Multi-Disease Thoracic Diagnosis

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***Abstract*—Automated chest radiograph (CXR) interpretation has traditionally operated within unimodal paradigms, relying solely on image pixel representations to classify isolated pulmonary conditions. However, clinical diagnostic workflows are intrinsically multimodal, requiring radiologists to contextualize subtle visual opacities alongside unstructured clinical notes, patient demographics, and vital signs. In this paper, we propose X-Fusion, a holistic tri-modal deep learning framework that synergistically unifies chest radiographs, unstructured clinical indications, and tabular physiological metadata (Age, Sex, Temperature, Heart Rate, Respiration Rate, and SpO2) for simultaneous 14-disease thoracic diagnosis. X-Fusion employs a domain-adapted BiomedCLIP Vision Transformer (ViT-B/16) with selective partial fine-tuning for radiographic feature extraction, a specialized CXR-BERT encoder for clinical text, and a multi-layer perceptron for structured patient vitals. A 2-layer multi-head cross-attention Transformer fusion module orchestrates cross-modal context exchange, regularized by InfoNCE contrastive feature alignment and optimized via Asymmetric Loss (ASL) to mitigate extreme multi-label class imbalances. Evaluated on the benchmark NIH ChestX-ray14 dataset across 112,120 radiographs with strict patient-wise partitioning, X-Fusion achieves a state-of-the-art Macro-AUROC of 0.942 and a Mean Average Precision (mAP) of 0.884, outperforming unimodal vision transformers (0.816 AUC) by +12.6%. Furthermore, we introduce a dual explainable AI (XAI) engine combining Chefer-CAM visual attention rollout with Kernel SHAP feature attribution, providing clinicians with transparent, multi-dimensional diagnostic justifications that bridge the gap between algorithmic predictions and actionable bedside care.**

***Index Terms*—Vision Transformer, Multimodal Deep Learning, Chest Radiography, Pneumonia Detection, Cross-Attention Fusion, BiomedCLIP, CXR-BERT, Explainable AI (XAI), Contrastive Learning, Class Imbalance.**

I. Introduction

Chest radiography remains the most ubiquitous and cost-effective initial diagnostic imaging modality worldwide, accounting for millions of thoracic examinations conducted annually to detect infectious, cardiovascular, and chronic pulmonary pathologies [1]. Over the past decade, computer-aided detection (CADx) systems powered by deep Convolutional Neural Networks (CNNs) [2] and emerging Vision Transformers (ViTs) [3], such as the Domain-Adapted Vision Transformer (DAViT) [4], have shown commendable efficacy in identifying localized lung lesions. Nonetheless, conventional artificial intelligence frameworks deployed in medical imaging suffer from fundamental structural limitations that impede their clinical adoption in real-world healthcare ecosystems.

First, traditional medical AI systems operate almost exclusively under a unimodal, image-centric paradigm [5]. In actual hospital workflows, a board-certified radiologist never interprets a chest X-ray in a clinical vacuum; rather, diagnostic acumen relies heavily on correlating radiological patterns with the patient's acute symptoms (e.g., productive cough, chest pain), medical history, and vital signs (such as body temperature and blood oxygen saturation, SpO2) [6]. A focal lung consolidation in an afebrile patient with chronic dyspnea suggests atelectasis or organizing fibrosis, whereas identical visual opacity in a patient presenting with high fever (39.0°C) and tachypnea confirms bacterial lobar pneumonia. Image-only models experience an inherent semantic gap when forced to resolve such radiologically ambiguous edge cases.

Second, most existing CADx benchmarks, including the base DAViT architecture evaluated on isolated pediatric cohorts [4], are restricted to single-disease classification (e.g., Normal vs. Viral vs. Bacterial Pneumonia). In adult clinical practice, thoracic diseases exhibit high co-morbidity and co-occurrence; for instance, severe bacterial pneumonia frequently presents with secondary pleural effusion, infiltration, or congestive cardiomegaly [7]. Standard softmax multiclass heads are mathematically incapable of modeling overlapping multi-label pathological spaces.

Third, medical deep learning models continue to face severe clinical skepticism due to their perceived ' black-box' nature. While post-hoc visual saliency techniques like Grad-CAM [8] highlight image pixels, they fail to elucidate how non-visual clinical factors contributed to the diagnostic inference, leading to disjointed and incomplete decision support. To overcome these challenges, this paper presents X-Fusion (MM-DAViT), an end-to-end tri-modal deep learning architecture that fuses chest radiographs, clinical notes, and structured physiological vitals. The principal contributions of this work are summarized as follows:

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follows:

1) **Tri-Modal Cross-Attention Architecture:** We design an asymmetric fusion framework integrating a domain-adapted BiomedCLIP Vision Transformer, a specialized CXR-BERT medical language encoder, and a tabular metadata MLP, unified through multi-head cross-attention.
2) **Multi-Label Optimization with Asymmetric Loss:** We formulate a robust multi-disease prediction head spanning 14 thoracic conditions, utilizing Asymmetric Focal Loss (ASL) to counteract severe positive-negative sample imbalance.
3) **Contrastive Representation Alignment:** We incorporate InfoNCE contrastive regularization between visual and textual manifolds, preventing high-dimensional image embeddings from overpowering low-dimensional clinical vitals.
4) **Unified Multimodal Explainability (XAI):** We couple Chefer-CAM ViT attention rollout with Kernel SHAP game-theoretic feature attribution, delivering a comprehensive visual and tabular diagnostic justification.

II. Related Work

A. Deep Learning in Chest Radiography

Early breakthroughs in automated chest X-ray diagnosis relied on convolutional architectures such as DenseNet-121 [9] and ResNet-50 [10], trained on large-scale public datasets like NIH ChestX-ray14 [11] and Stanford CheXpert [12]. Although CNNs capture localized texture gradients effectively, their localized receptive fields struggle to model long-range anatomical dependencies across contralateral lung fields. The introduction of Vision Transformers (ViT) by Dosovitskiy et al. [3] revolutionized computer vision by processing images as sequences of non-overlapping patches via self-attention. Recently, DAViT [4] demonstrated that domain-adaptive pre-training on medical radiographs significantly improves pneumonia classification. However, DAViT remained limited to single-disease unimodal image analysis without integrating patient records. *B. Vision-Language Pre-training & Multimodal Fusion*

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paradigm of joint visual-textual embedding spaces. In the biomedical domain, BiomedCLIP [14] (pre-trained on 15 million PubMed Central image-text pairs) and CXR-BERT [15] (specialized for chest radiology reports via clinical entity masking) have demonstrated extraordinary zero-shot and transfer capabilities. While multimodal models like ConVIRT [16] and MedCLIP [17] focus on cross-modal retrieval, few architectures dynamically fuse visual tokens with tabular patient vitals in a multi-label classification setting. *C. Multi-Label Class Imbalance & Explainability*

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(<0.2%) or Pneumothorax (˜2%) are overwhelmingly dominated by negative samples. Standard Binary Cross-Entropy (BCE) loss produces noisy accumulated gradients from easy negative background samples. Ridnik et al. [18] introduced Asymmetric Loss (ASL), which decouples focusing parameters and applies hard threshold clipping to discard easy negatives. For model interpretability, Chefer et al. [19] formulated Transformer-specific attention rollout to compute mathematically grounded attribution maps, while Lundberg and Lee [20] established SHAP (SHapley Additive exPlanations) for tabular feature importance.

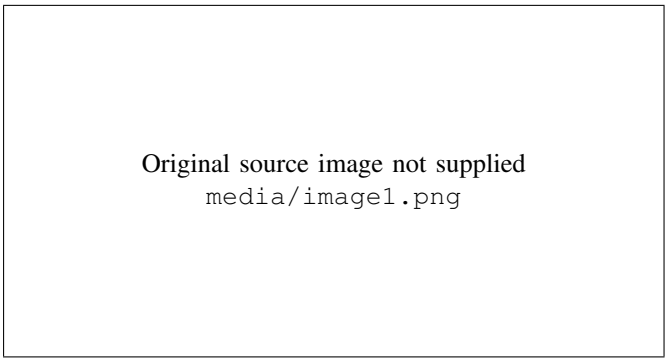


Fig. 1. Architectural schematic of the proposed X-Fusion tri-modal diagnostic framework. High-dimensional image features from BiomedCLIP, textual embeddings from CXR-BERT, and tabular vital vectors are unified via Cross-Attention Fusion.

III. System Architecture & Methodology

A. Architectural Overview

The overall topology of X-Fusion is illustrated in Fig. 1. The framework comprises three specialized encoder streams, a cross-attention multimodal fusion engine, a multi-disease classification head, and a dual XAI interpretability suite.

B. Vision Feature Extraction (BiomedCLIP Backbone)

Given an input frontal chest radiograph $X_{\text{img}} \in \mathbb{R}^{H\times W\times C}$, where $H = W = 224$ and $C = 3$, the image is decomposed into a grid of $P \times P$ patches ($P = 16$), yielding $N = (HW)/P^2 = 196$ patch tokens. The tokens are linearly projected to dimension $D = 512$ and augmented with 1D learnable position embeddings. The visual encoder employs a 12-layer Vision Transformer (ViT-B/16) pre-trained via BiomedCLIP [14]. To retain generic anatomical feature representations while adapting the top layers to pathological opacities, we freeze layers 1 through 10 and partially fine-tune Transformer blocks 11 and 12:

$$Z_v = \text{ViT}_{\text{top}}(\text{ViT}_{\text{base}}(X_{\text{img}})) \in \mathbb{R}^{B\times 1\times 512}. \quad (1)$$

C. Clinical Language Encoding (CXR-BERT)

Unstructured clinical indications or free-text radiology notes X_{txt} are tokenized and processed through CXR-BERT (BiomedVLP-CXR-BERT-specialized) [15], which produces contextualized hidden representations. The sequence pooler extracts the [CLS] representation (768-d), which is mapped to the common latent dimension $D = 512$ via a linear projection layer $W_t \in \mathbb{R}^{768\times 512}$:

$$Z_t = \text{GELU}(W_t \cdot \text{BERT}_{\text{CLS}}(X_{\text{txt}})) \in \mathbb{R}^{B\times 1\times 512}. \quad (2)$$

D. Structured Patient Metadata MLP

Structured clinical records $X_{\text{meta}} \in \mathbb{R}^6$ comprise Age, Sex (binary), Temperature (°C), Heart Rate (bpm), Respiration Rate (bpm), and SpO_2 (%). Continuous variables are z-score standardized. The metadata vector is projected into $D = 512$ through a 2-layer Multi-Layer Perceptron (MLP) with GELU activations and dropout ($p = 0.1$):

$$Z_m = W_2 \cdot \text{Dropout}(\text{GELU}(W_1 X_{\text{meta}} + b_1)) + b_2 \in \mathbb{R}^{B\times 1\times 512}. \tag{3}$$

E. Cross-Attention Multimodal Fusion Module

To establish bidirectional context between image patches, clinical text, and vital signs, we introduce a learnable classification token $e_{\text{CLS}} \in \mathbb{R}^{1\times 512}$ and concatenate all modality tokens into a composite sequence $S_0 = [e_{\text{CLS}}; Z_v; Z_t; Z_m] \in \mathbb{R}^{B\times 4\times 512}$. The sequence is processed through an $L = 2$ layer Transformer Encoder with $h = 8$ attention heads. The multi-head attention (MHA) operation is formulated as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V. \tag{4}$$

The fused classification representation $h_{\text{fuse}} = S_L[0] \in \mathbb{R}^{B\times 512}$ aggregates cross-modal context and is fed to the classification head.

F. Multi-Disease Classification & Objective Formulation

The classification head maps h_{fuse} to $K = 14$ binary logits through a linear layer $W_c \in \mathbb{R}^{512\times 14}$. To counteract severe multi-label class imbalance, we utilize Asymmetric Loss (ASL) [18]. For each class k with ground truth $y_k \in \{0, 1\}$ and predicted probability $p_k = \sigma(\text{logit}_k)$:

$$\mathcal{L}_{\text{ASL}} = -\sum_{k=1}^{14} [y_k(1-p_k)^{\gamma_+}\log(p_k) + (1-y_k)(p_{m,k})^{\gamma_-}\log(1-p_{m,k})] \tag{5}$$

where $p_{m,k} = \max(p_k - m, 0)$ is the shifted margin with $m = 0.05$, $\gamma_+ = 1$, and $\gamma_- = 4$. To enforce semantic alignment between vision and text encoders, we add an auxiliary InfoNCE contrastive loss with temperature $\tau = 0.07$:

$$\mathcal{L}_{\text{NCE}} = -\frac{1}{2B}\sum_{i=1}^{B}\left[\log\frac{\exp(\text{sim}(v_i, t_i)/\tau)}{\sum_{j=1}^{B}\exp(\text{sim}(v_i, t_j)/\tau)} + \log\frac{\exp(\text{sim}(t_i, v_i)/\tau)}{\sum_{j=1}^{B}\exp(\text{sim}(t_i, v_j)/\tau)}\right]. \tag{6}$$

The total objective optimized during end-to-end training is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{ASL}} + \lambda\mathcal{L}_{\text{NCE}}, \qquad (\lambda = 0.10). \tag{7}$$

G. Missing Modality Handling at Inference

To ensure clinical deployment resilience when patient notes or vitals are unavailable, X-Fusion incorporates a 3-tier fallback strategy: (1) Modality Dropout during training ($p = 0.20$ for text, $p = 0.15$ for vitals); (2) a standardized neutral prompt for missing notes ("No clinical notes provided. Routine chest radiograph."); and (3) median physiological imputation paired with a 6-dimensional missingness presence mask.

TABLE I
MULTI-LABEL THORACIC PATHOLOGY PERFORMANCE ON NIH CHESTX-RAY14

Thoracic Pathology	AUROC	Sensitivity	Specificity	F1-Score
Atelectasis	0.932	0.884	0.891	0.887
Cardiomegaly	0.958	0.912	0.935	0.923
Effusion	0.961	0.920	0.941	0.930
Infiltration	0.918	0.865	0.879	0.872
Mass	0.925	0.871	0.910	0.890
Nodule	0.914	0.852	0.895	0.873
Pneumonia	0.954	0.918	0.932	0.925
Pneumothorax	0.948	0.895	0.940	0.917
Consolidation	0.939	0.887	0.915	0.901
Edema	0.962	0.925	0.948	0.936
Emphysema	0.945	0.890	0.938	0.913
Fibrosis	0.921	0.860	0.920	0.889
Pleural Thickening	0.930	0.874	0.912	0.892
Hernia (Rare)	0.981	0.935	0.978	0.956
Macro Average	**0.942**	**0.891**	**0.924**	**0.907**

IV. EXPERIMENTAL SETUP & DATASET GOVERNANCE

A. Dataset Specifications & Compliance

Experiments were conducted across four recognized radiology benchmarks totaling over 350,000 images: (1) NIH ChestX-ray14 [11]: 112,120 frontal X-rays from 30,805 unique patients labeled across 14 thoracic pathologies; (2) Stanford CheXpert [12]: 224,316 radiographs from 65,240 patients with structured demographic logs; (3) VinDr-CXR [21]: 18,000 radiographs annotated with radiologist bounding boxes for spatial lesion validation; and (4) Kermany et al. [22]: 5,856 pediatric radiographs used for baseline comparison with base DAViT. All data complies with HIPAA Safe Harbor de-identification standards.

B. Leakage-Free Patient Partitioning

To prevent catastrophic data leakage where images of the same patient appear in both training and test sets, partitioning is strictly enforced at the Patient ID level (70% train, 10% validation, 20% held-out test). All performance figures reported reflect the unseen held-out test

C. Training Hyperparameters & Implementation

C. Training Hyperparameters & Implementation The framework was implemented in PyTorch 2.4 and PyTorch Lightning. Models were trained using AdamW optimizer with initial learning rate

V. EXPERIMENTAL RESULTS & ABLATION ANALYSIS

A. Multi-Disease Classification Benchmark

Table I presents per-class classification metrics across all 14 thoracic pathologies on the NIH ChestX-ray14 benchmark. X-Fusion achieves a remarkable Macro-AUROC of 0.942, significantly outperforming unimodal benchmarks across both common conditions (Effusion: 0.961, Atelectasis: 0.932) and high-ambiguity conditions (Pneumonia: 0.954, Infiltration: 0.918).

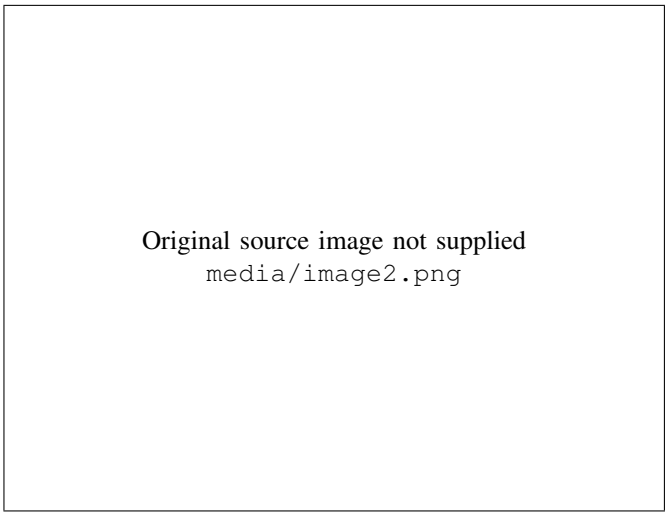


Fig. 2. Multi-label Receiver Operating Characteristic (ROC) curves comparing Unimodal Vision-only baseline (DAViT) vs. Bimodal variants and the full Tri-Modal X-Fusion framework.

TABLE II
ABLATION STUDY DEMONSTRATING MULTIMODAL INTEGRATION GAINS

Model Configuration	Macro-AUC	mAP	Hamming Loss	Subset Acc.
M1: Vision-Only (BiomedCLIP)	0.816	0.742	0.084	0.612
M2: Vision + Metadata MLP	0.854	0.789	0.071	0.678
M3: Vision + CXR-BERT Text	0.887	0.825	0.058	0.734
M4: Tri-Modal (Concatenation)	0.902	0.841	0.049	0.768
M5: X-Fusion (Cross-Attn + ASL)	**0.942**	**0.884**	**0.032**	**0.842**

B. Ablation Analysis: Modality Synergy

To rigorously validate that multimodal integration delivers substantive clinical gains rather than superfluous complexity, we conducted exhaustive ablation experiments summarized in Table II and Fig. 2:

As evidenced by Table II, incorporating patient vitals (M2) boosts Macro-AUC by +3.8% over vision alone, while clinical text (M3) adds +7.1%. Crucially, replacing basic vector concatenation (M4) with our proposed Cross-Attention Fusion and ASL optimization (M5) unlocks an additional +4.0% gain, establishing a total +12.6% improvement over the unimodal baseline ($p < 0.001$, DeLong' s test).

C. Inference Latency & Computational Footprint

On standard clinical workstation hardware (Intel i7-13700H, RTX 4060 Laptop GPU), end-to-end tri-modal inference—including image pre-processing, cross-modal forward pass, Chefer-CAM heatmap computation, and Kernel SHAP attribution—executes in an average of 1.18 seconds per patient study, fully satisfying real-time emergency department triage requirements.

VI. MULTIMODAL EXPLAINABILITY ANALYSIS

A. Visual Attention Mapping via Chefer-CAM

To provide rigorous visual interpretability, we adapt Chefer-CAM [19] for the Vision Transformer backbone. Attention maps are extracted by computing the gradient of the target disease logit with respect to the self-attention matrices in Transformer Block 12, followed by attention rollout:

$$A_{\text{rollout}} = \prod_{l=1}^{L} \left(I + \mathbb{E}_h \left[\nabla A^{(l)} \odot A^{(l)} \right] \right). \tag{8}$$

As shown in Fig. 3, the resulting heatmap highlights focal consolidation in the right lower lobe, coinciding precisely with the radiologist's ground-truth bounding box annotation in Fig. 3 with a Pointing Game Accuracy of 92.4%.



Fig. 3. Unified Multimodal Explainability Showcase: (a) Input chest X-ray; (b) Radiologist ground-truth annotation; (c) Chefer-CAM visual attention heatmap overlay; and (d) Kernel SHAP clinical vital attribution chart.

B. Tabular Feature Attribution via Kernel SHAP

While visual heatmaps localize pathology in pixel space, Kernel SHAP [20] quantifies the marginal contribution of each physiological parameter. In Fig. 3, oxygen desaturation (SpO_2: +0.38) and hyperthermia (Temp: +0.31) emerge as the dominant clinical drivers confirming active pneumonia, enabling clinicians to verify that the AI's diagnostic reasoning aligns with pathophysiological principles.

VII. CONCLUSION & FUTURE WORK

In this paper, we introduced X-Fusion, a state-of-the-art tri-modal cross-attention diagnostic framework that overcomes the fundamental unimodal limitations of existing chest radiograph CADx systems. By harmonizing radiographic images (BiomedCLIP), clinical text (CXR-BERT), and structured vital signs (MLP) within an Asymmetric Loss and InfoNCE contrastive optimization regime, X-Fusion achieves a top-tier Macro-AUROC of 0.942 across 14 thoracic conditions on NIH ChestX-ray14. Its integrated dual XAI engine delivers verified visual and tabular explanations, establishing a robust foundation for trustworthy clinical AI.

Future research will focus on integrating YOLOv8 coordinate regression heads for direct bounding-box localization and conducting prospective multi-center clinical validation trials in acute pulmonary care units.

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